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https://tome.app/startec-2cc/final-presentation-clxmdpduu0fezus2scdyh20as

STARTEC

SPACE TRASH AVOIDANCE AND
REMOVAL TECHNOLOGY ECOSYSTEM

Airbus Café

HSB [2024-07-04 T15:20:23Z] ECOMAT

WHAT WE DO

MISSION STATEMENT

"To ensure the **sustainability and safety of space operations** by providing a comprehensive solution for the tracking, removal and prevention of space debris."

USE CASES

Space debris poses a significant collision risk to operational satellites

Detection, tracking, and collision alert

Thousands of big and millions of small debris in space

Capture and removal debris <10cm through controlled deorbiting

STARTEC wants to ensure the protection and uninterrupted operation of satellites and to preserve space for future generations

How ?

MISSION REQUIREMENTS

Detection/Tracking

Advanced Tracking-System capable of identifying and monitoring debris of various sizes.

Capture

ADR-Satellites for reliable and versatile debris capturing and transporting.

Transfer

Transfer-Station for orbital transfer, transferring debris, refuelling and maintenance

Removal

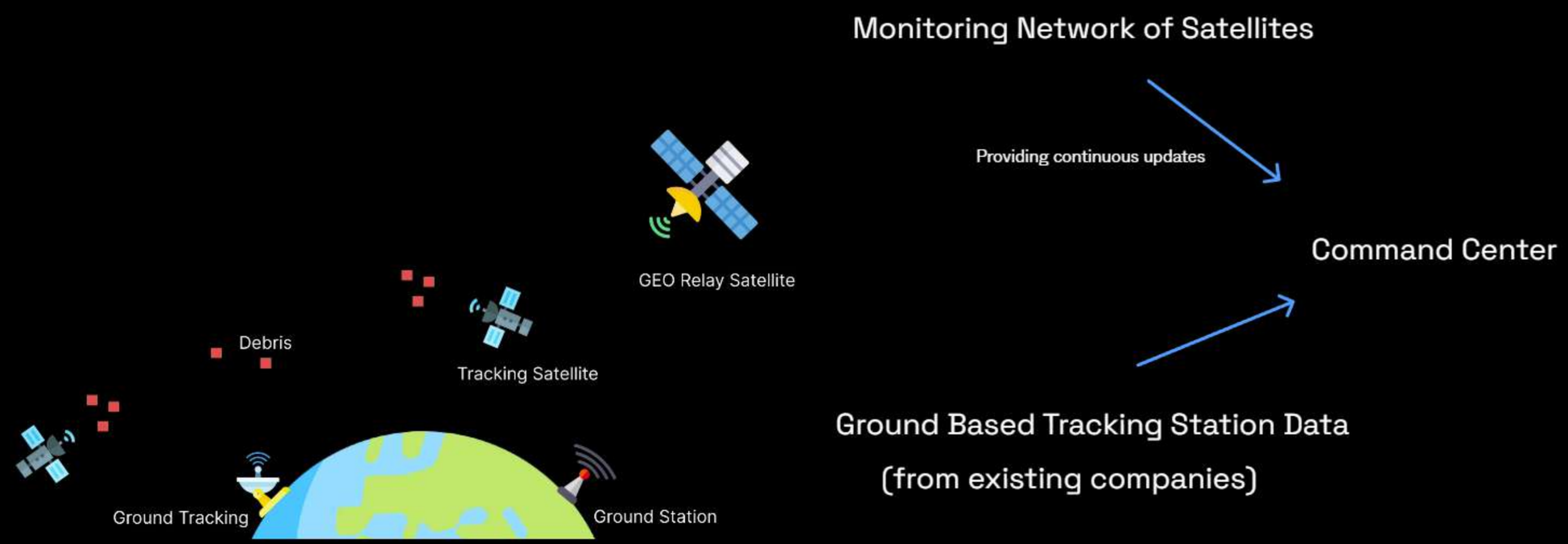
Deorbiting-System for executing controlled deorbit maneuvers to safely remove debris

Software

Integrated Software-Platform for data analysis, mission planning, and real-time monitoring.

OVERALL OPERATIONAL CONCEPT

Detection & Tracking

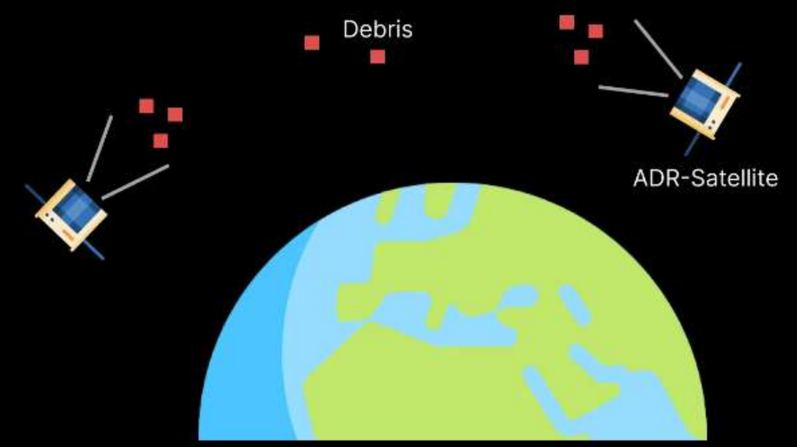


Capture & Collect

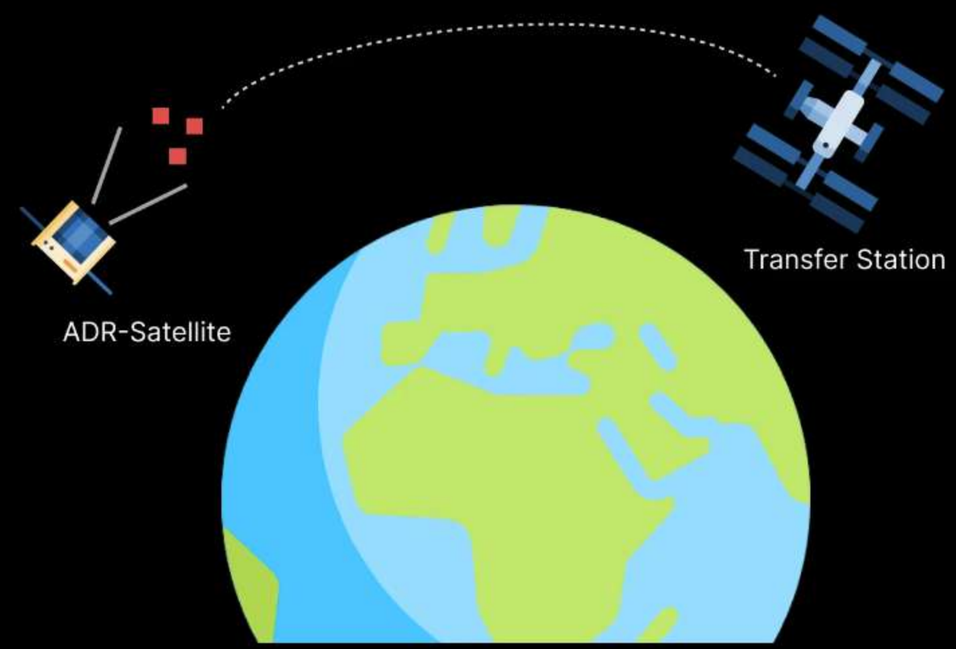
Equipped with
Capture System

ADR-Satellites

Equipped with Sensors to
navigate and target debris



Transfer



ADR-Satellites

Transport captured debris back to

Transfer Station

Transfer/Store debris via robotic arm

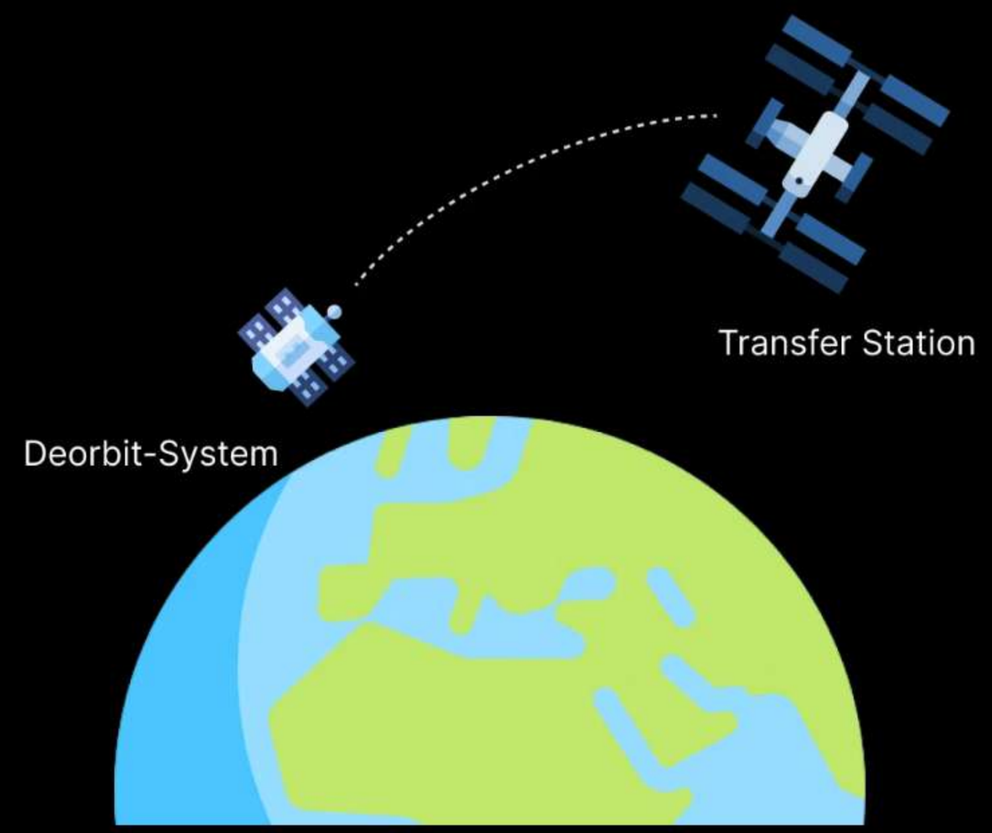
Removal

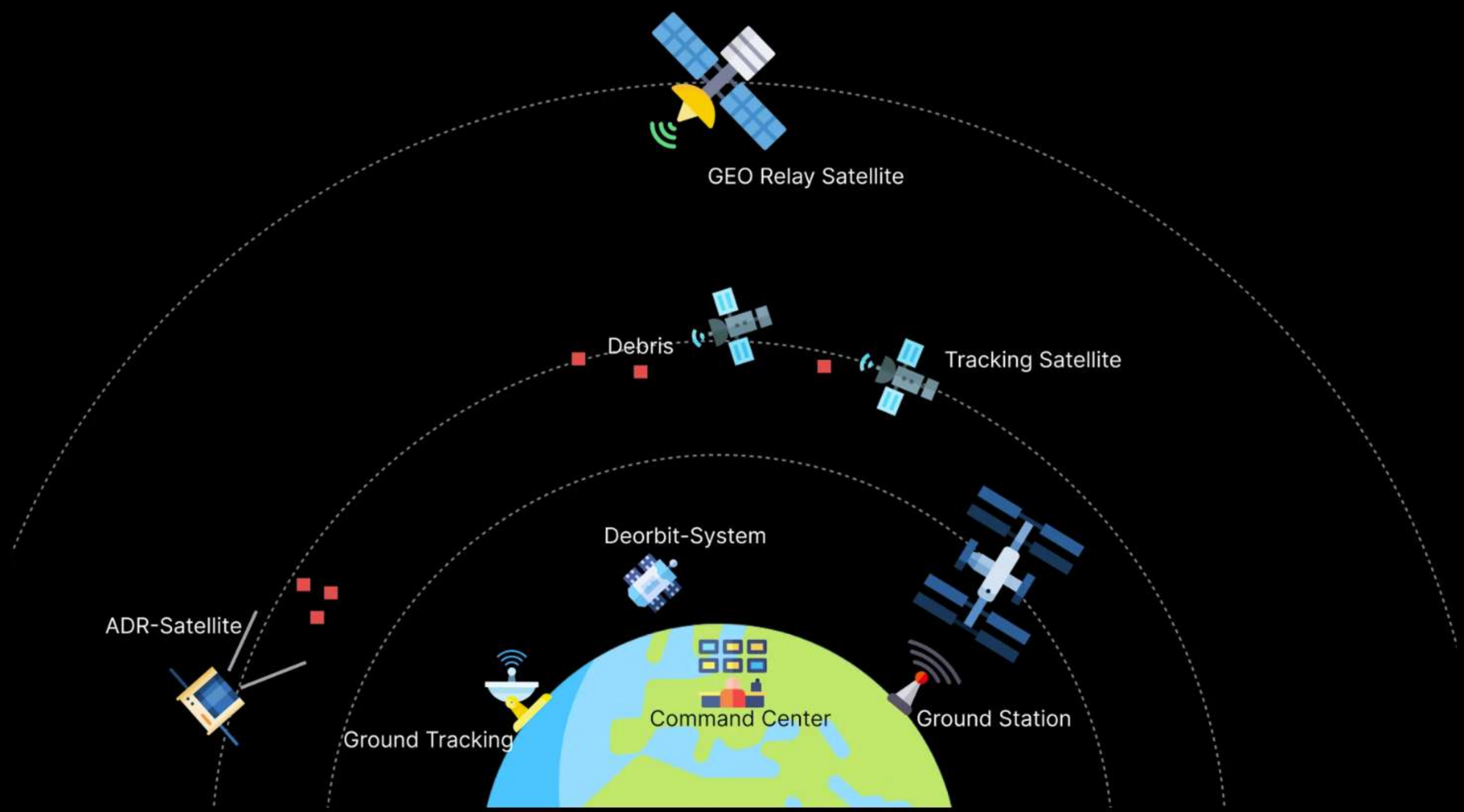
Main Station

Transfer of debris via robotic arm to

Deorbit-System

Robust Propulsion System for controlled atmospheric rentry

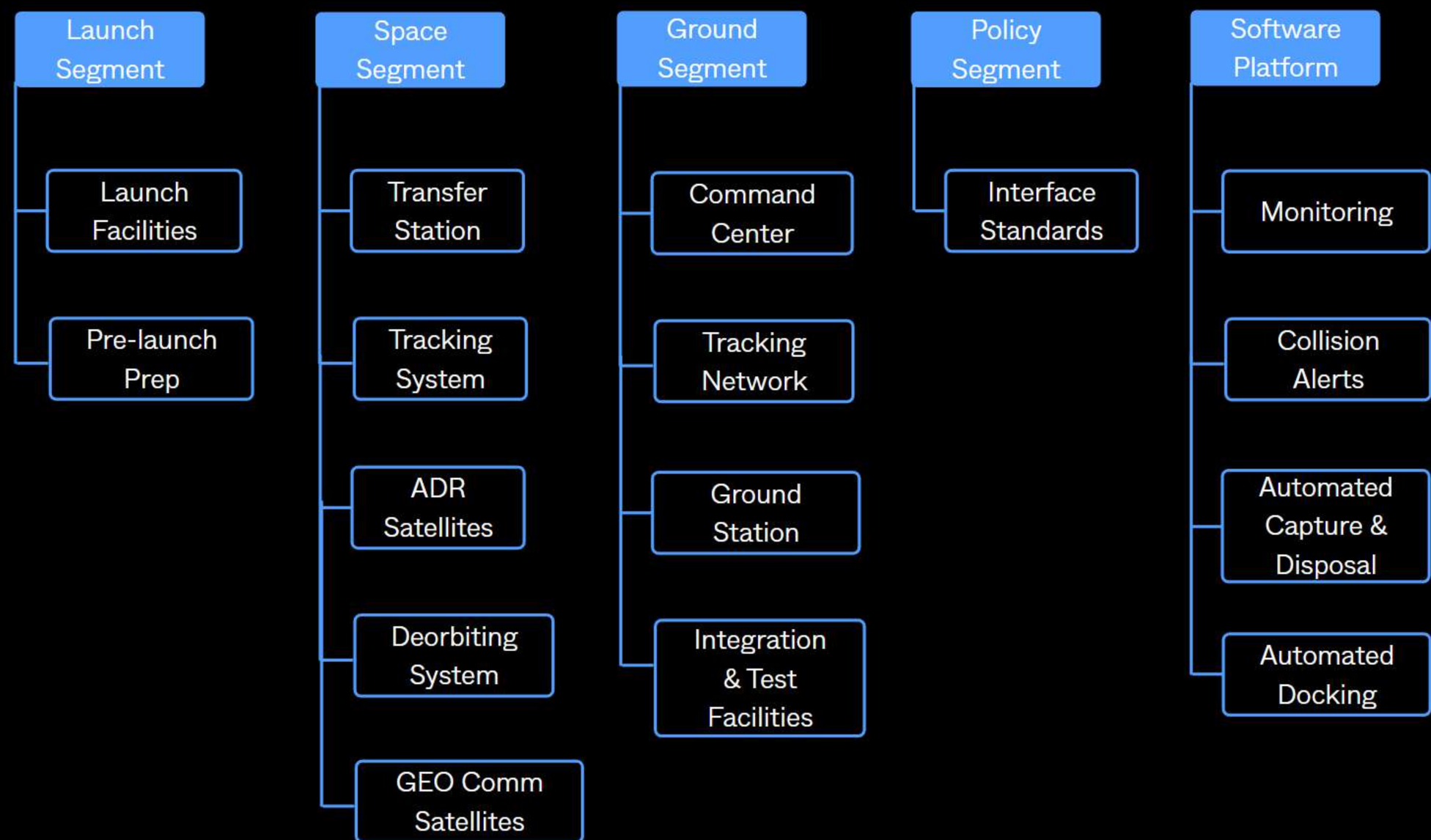



GEO Relay Satellite
Debris
Tracking Satellite
Deorbit-System
ADR-Satellite
Ground Tracking
Command Center
Ground Station

The Blueprint

SYSTEM DESIGN

SYSTEM BREAKDOWN



MONITORING NETWORK

- Spaced Based Tracking Satellites
- Ground based tracking data from companies like Leolabs

Orbital Configuration

30

SATELLITES

97.4 °

INCLINATION (SSO)

5

PER PLANE

72°

PHASE SHIFT

700 Km

ORBITAL ALTITUDE

Comms

- Inter-satellite Links
- Communication with Ground Stations via GEO
- High-bandwidth Data Downlink

Sub-Systems

Propulsion: Hall effect Thrusters with Xenon Propellant

Sensor Suite: 3D Imaging Sensor, IR Sensor, LiDAR

On Board Data Processing: AI Algorithms for navigation and relative positioning

Power: Solar Panels, Batteries

ACS: Reaction Wheels, Star Trackers, Gyro, Sun Tracker

Size Range

1cm-10m

Distance Range

up to 400Km

ADR SATELLITES

Multi-mission reusable satellite consisting of

- Capture Module
- Service Module

Capture Module

1 m
DIAMETER

50 Kg
MASS

0.5 m
LENGTH

1-10 cm
DEBRIS SIZE RANGE

Service Module

Propulsion: Monoprop thrusters using Hydrazine

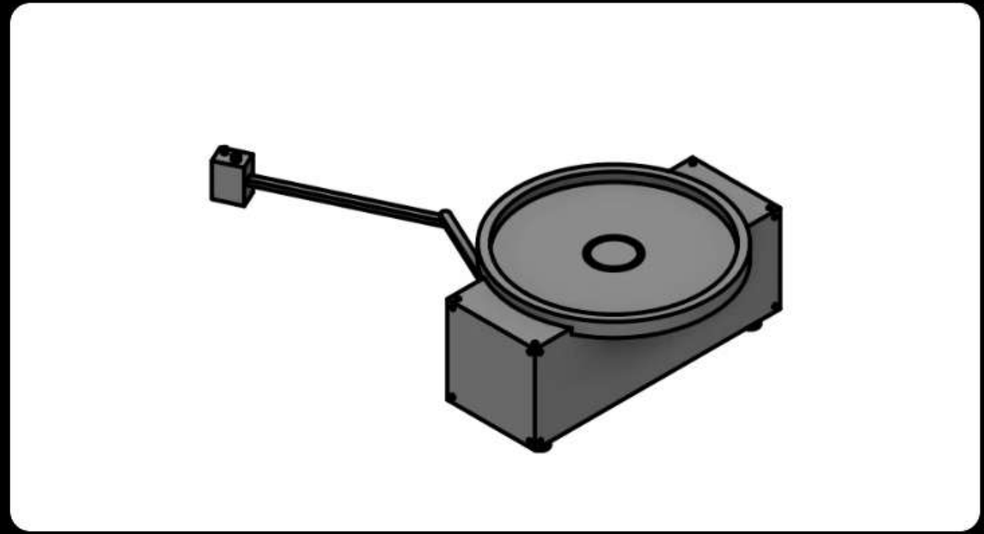
Sensor Suite: IR Sensor, LiDAR

On Board Data Processing: AI Algorithms for navigation and capture

AOCS: Reaction Wheels, Star Trackers, Gyro, Sun Tracker

Power: Solar Panels, Batteries

100 Kg
WET MASS



Structure

- Cylindrical Shell
- Central Cone
- Fibers

Operations Sequence

- **Deployment** — Navigation to target orbit
- **Approach** — Far field rendezvous using IR → Near field rendezvous using LiDAR
- **Capture** — Final Approach → Hatch Opening → Capture → Hatch Closing
- **Return** — Return maneuvring to transfer station

TRANSFER STATION

A Scalable Hub for

- Refuelling, Transferrring and Maintaining ADR Satellites
- Transfer of Capture Module from ADR Satellites to Deorbit Systems
- Docking of Deorbit System

Mass

15,700 Kg

Size

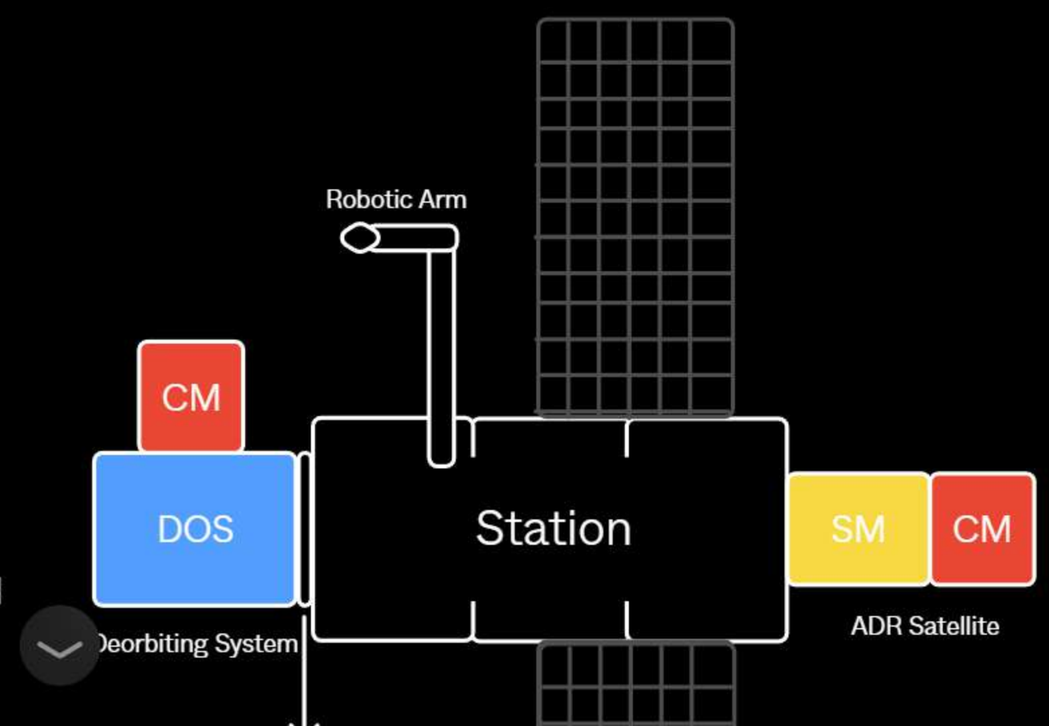
20 x 5 m

Power: Solar Arrays ~100 kW Power, Batteries

Propulsion: Hall effect Thrusters with Xenon Propellant

Robotic Arm: 10m with 7 degrees of freedom and end fixture

On Board Data Processing: AI Algorithms for



- Transfer of Capture Module from ADR Satellites to Deorbit Systems
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15,700 Kg

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Power: Solar Arrays ~100 kW Power, Batteries

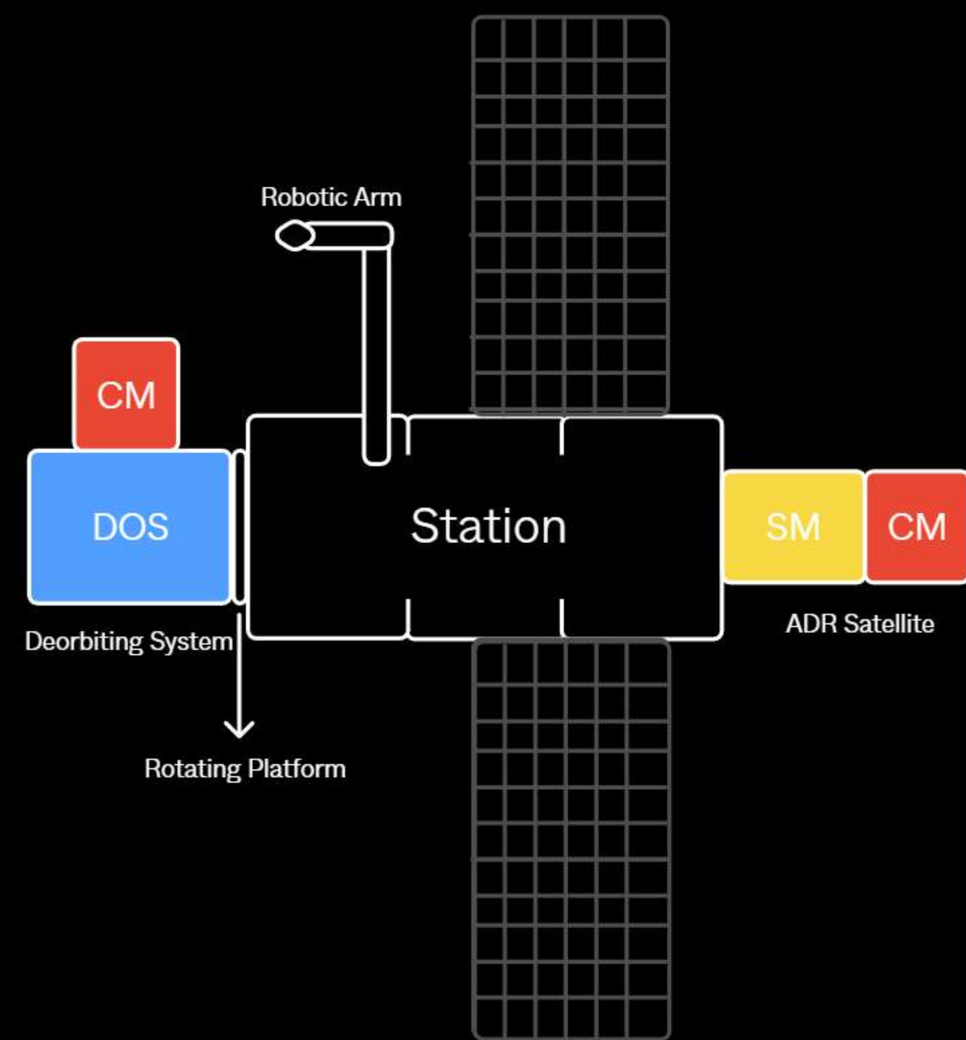
Propulsion: Hall effect Thrusters with Xenon Propellant

Robotic Arm: 10m with 7 degrees of freedom and end fixture

On Board Data Processing: AI Algorithms for navigation to target orbit, autonomous docking and station keeping

AOCS: Reaction Wheels, Star Trackers, Gyro, Sun Tracker, LiDAR, IR

Comms: Geo Relay



SM - Service Module
CM- Capture Module
DOS - Deorbit System

Operations Sequence

Deployment	Assembly in orbit
Orbital Maneuvering	Large Orbital Maneuvering → 50 Km below target orbit
Docking	ADR, Deorbit System, Capture Module Docking
Refuelling	Station & ADR Refuelling
Capture Module Transfer	Using Robotic Arm

SERVICER- REFUELLING SYSTEM

Provide propellant refuelling services to the STARTEC Transfer Station in orbit

ASTROSCALE (external provider)

Autonomous docking and Refuelling of the Station

Could potentially perform other supply and servicing operation

DEORBITING SYSTEM

Docked at the STARTEC Orbital Transfer Station, this platform features multiple docking ports for attaching capture modules transferred from ADR satellites.

A rotating plate mechanism facilitates efficient attachment of these modules using a robotic arm.

Power: Solar Arrays, Batteries

Propulsion: Hydrazine Monoprop engines

ACS: Reaction Wheels, Star Trackers, Gyro, Sun Tracker, IR

Comms; Geo Relay

Mass

1000 Kg

Size

5 x 2.5 m

Docking Ports

6

Operations Sequence

Module Attachment	Capture Module transfer from station
Deorbit Burn	High Thrust Burn to lower orbit
Atmospheric Reentry	Complete incineration of captured debris and module components

A Comprehensive Solution for Space debris.

SOFTWARE PLATFORM

Real-Time Tracking

- Satellite Positioning
- Live Updates

Collision Avoidance

- Alert System
- Maneuver Planning

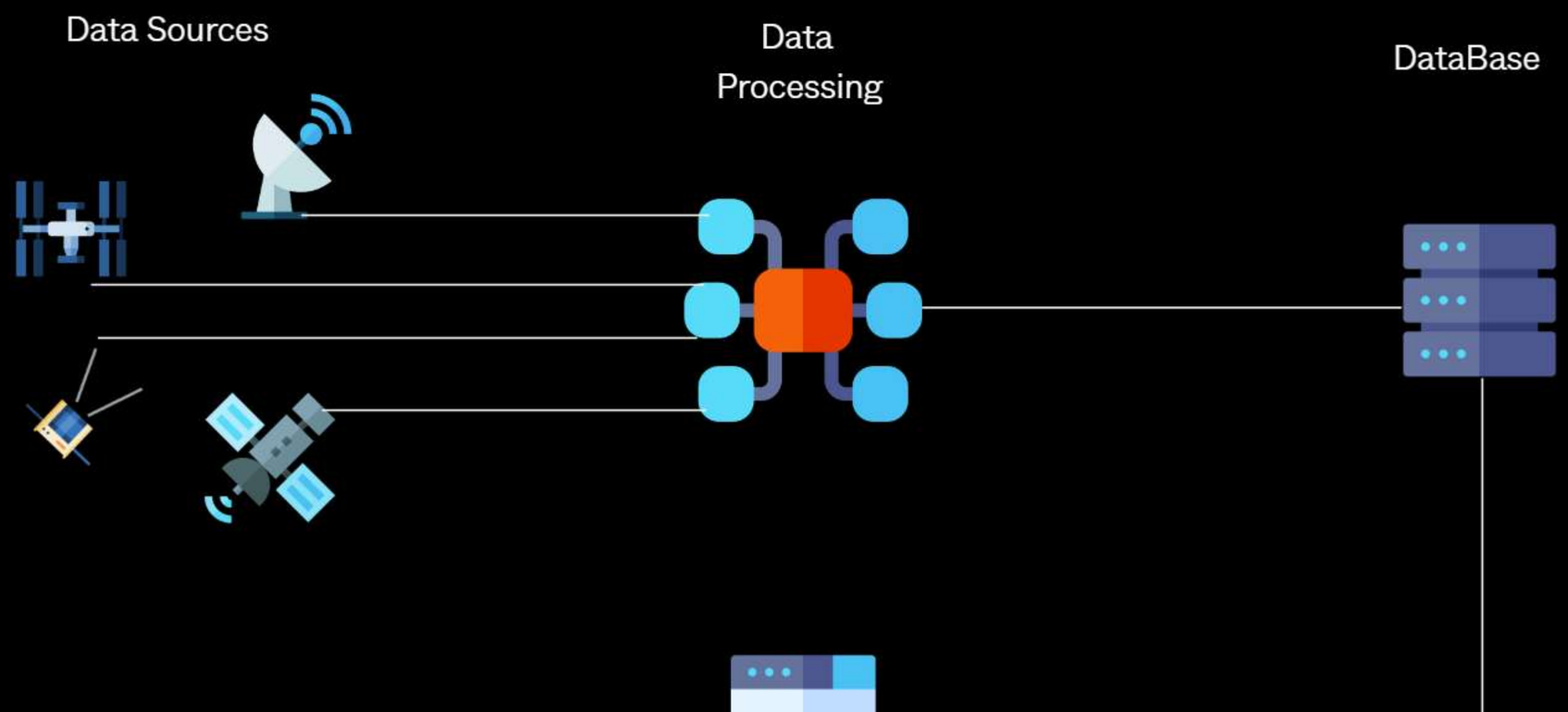
Mission Management

- Operational Control
- Resource Allocation

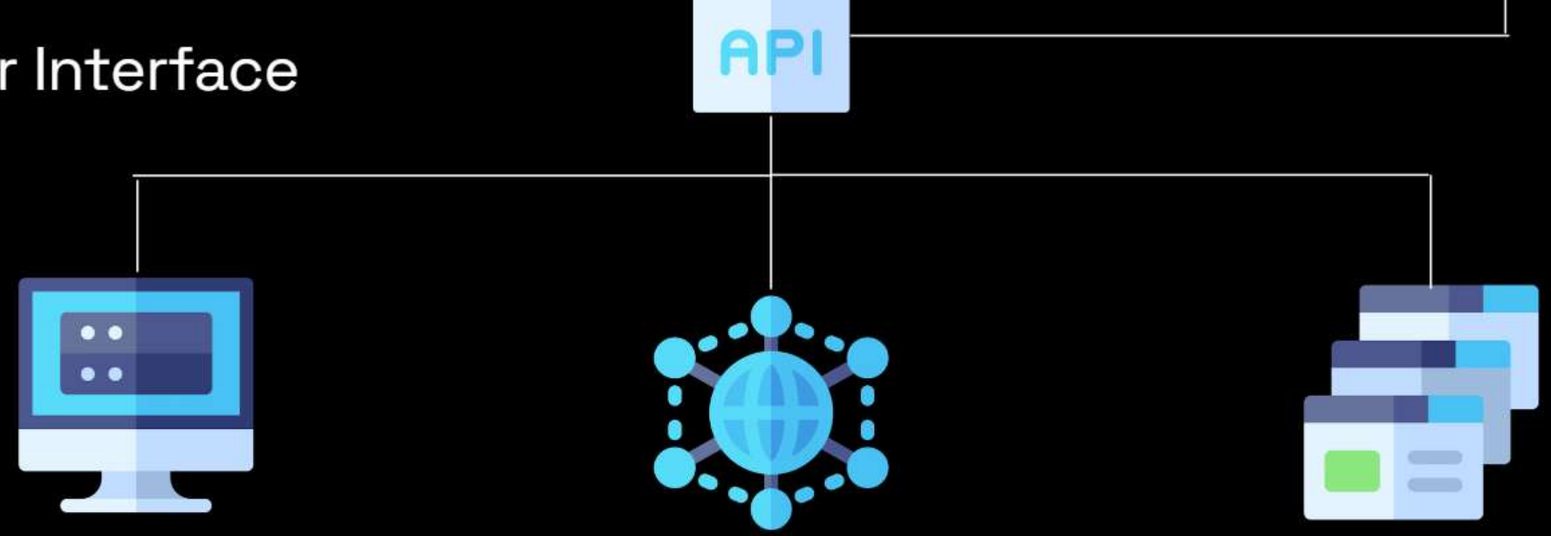
User Interaction

- Customer Interface
- Public Access

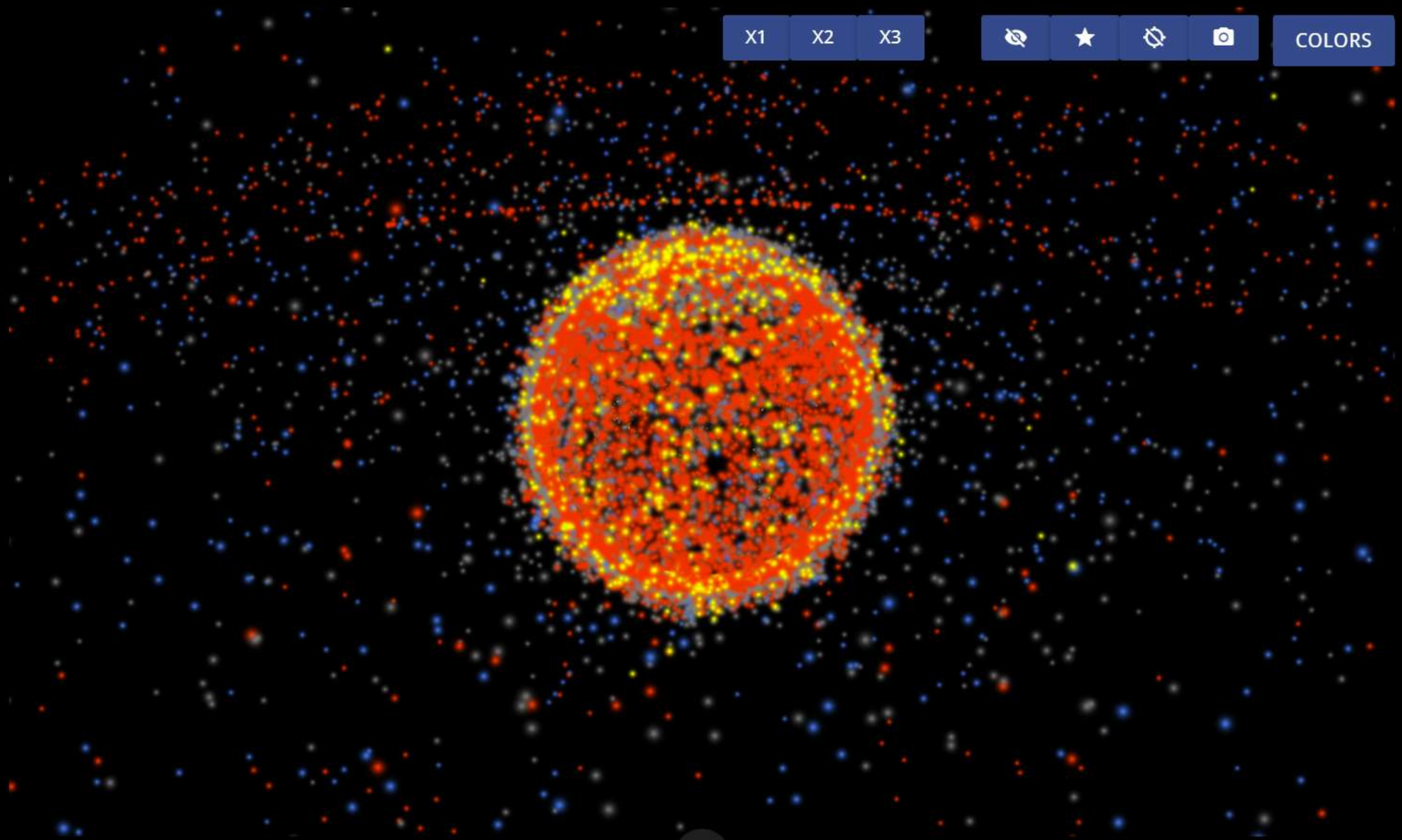
Centralized Data Integration



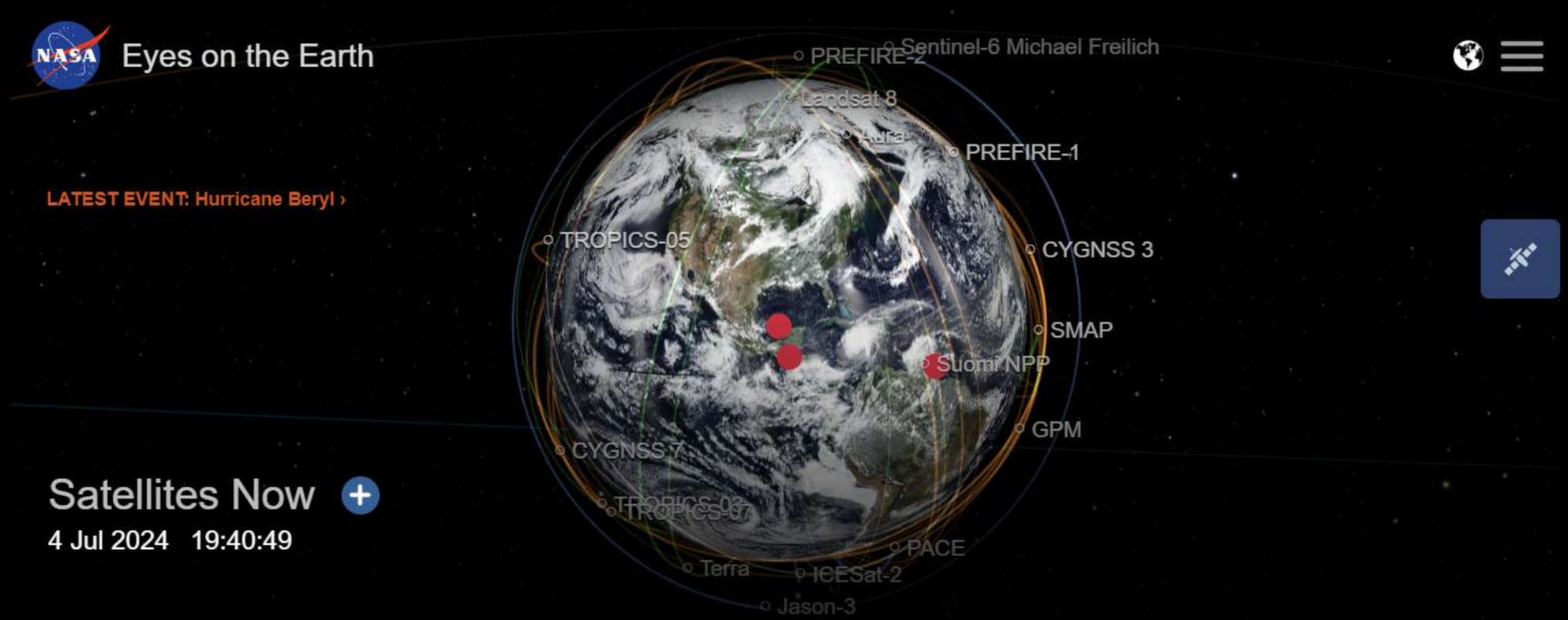
User Interface



3D Visualization

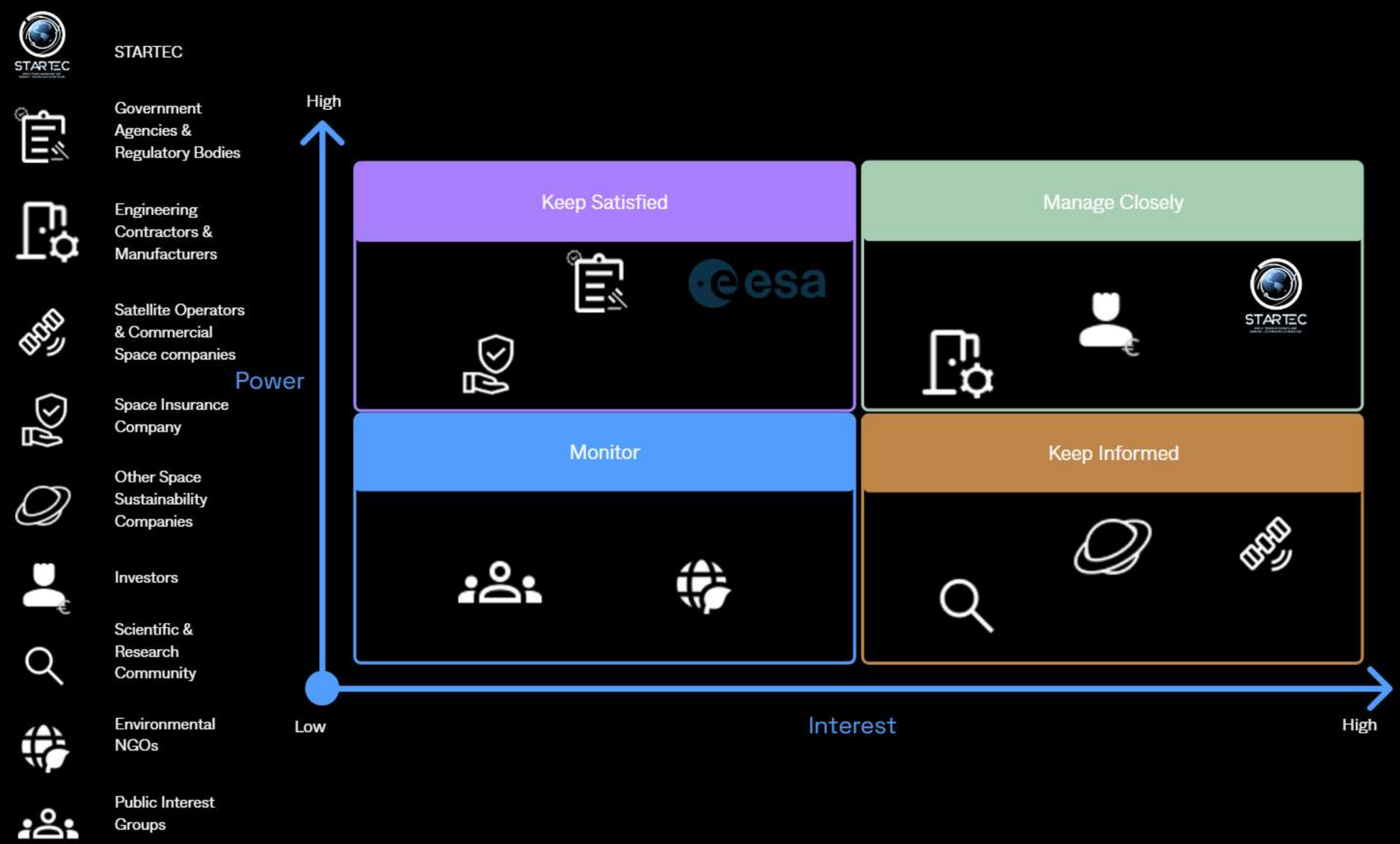


3D Visualization - NASA



KEY PLAYERS

STAKEHOLDERS



LEGAL CONSTRAINTS

Essential

do not create more space debris in our pursuit to reduce space debris

- ESA's Space Debris Mitigation Requirements
- ISO 24113: international Space Debris Mitigation Requirements

General

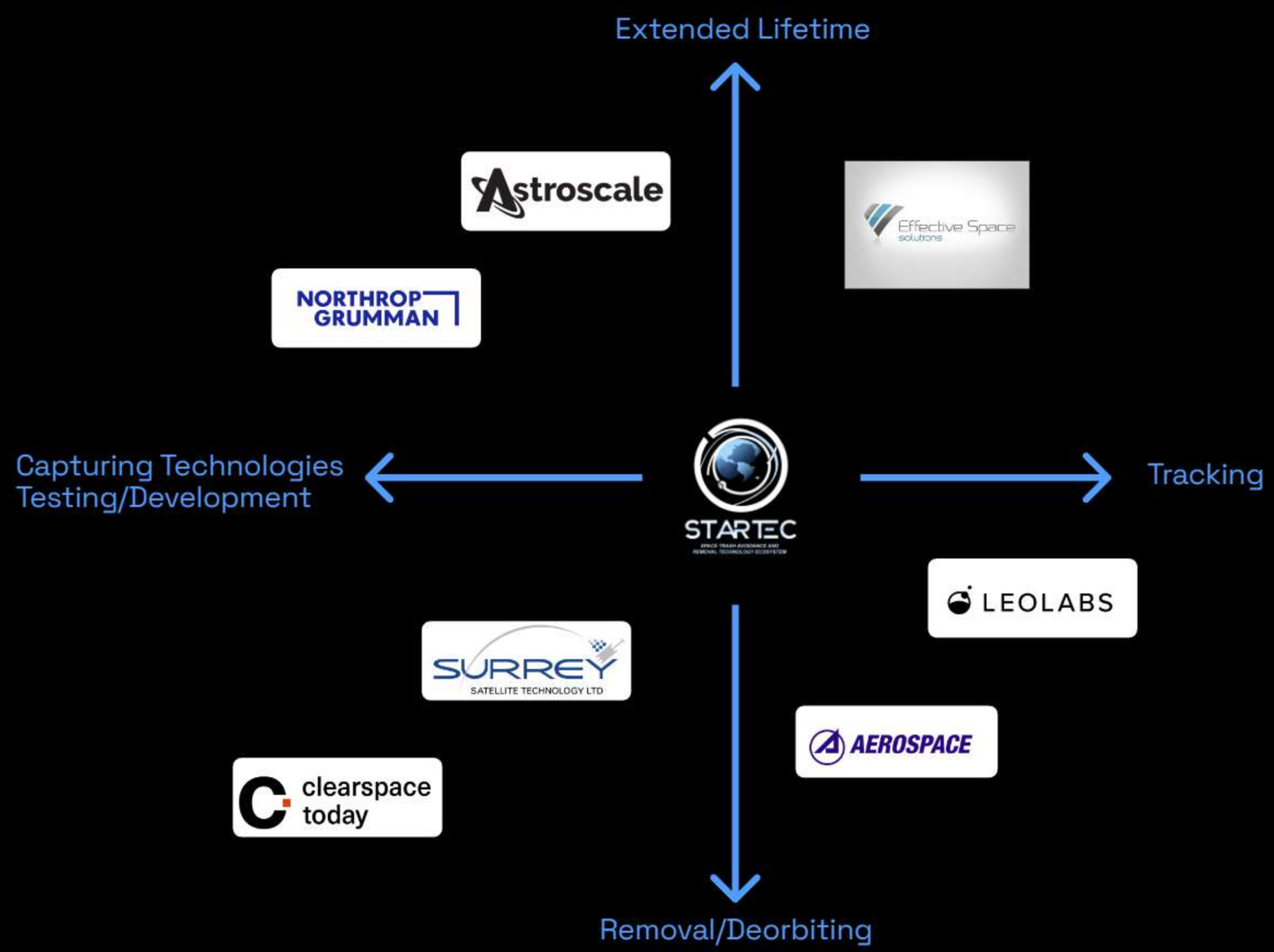
- Outer Space Treaty 1967
- Satellite Data Security Act 2008
- UNOOS's Space30 Agenda

Our Contracts

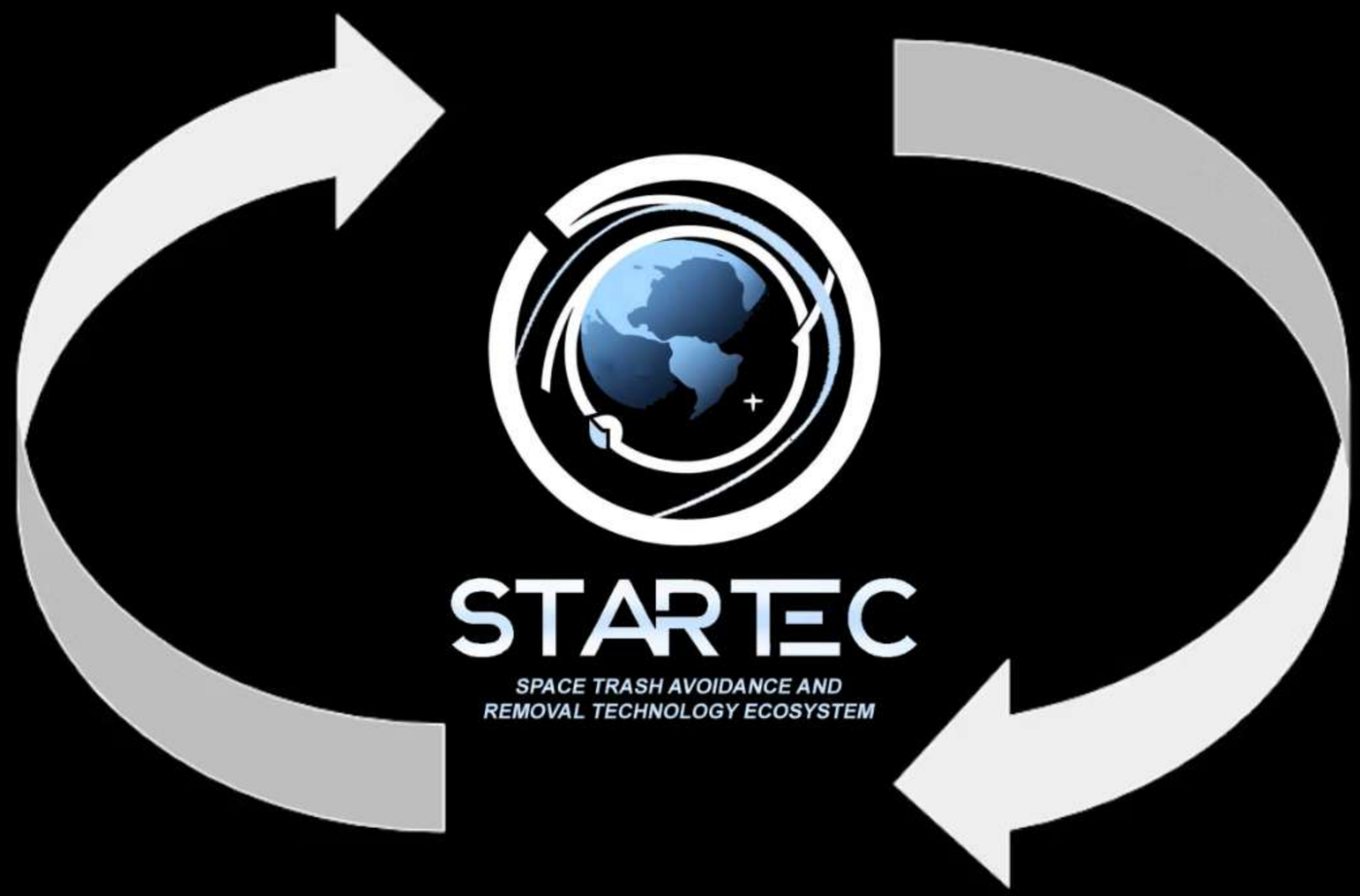
- Safety requirements of Insurance company
- Data sharing agreements
- Engineering & Manufacturing contracts

BUSINESS SIDE

THE COMPETITION



Removal/Deorbiting



We bring all of it together!

FINANCES

Category	RemoveDEBRIS	STARTEC	NOTES
Design and Engineering	$2,000,000	$2,000,000	Similar initial design costs, scaled for larger project scope.
Hardware Development	$6,000,000	$15,000,000	Higher due to 30 satellites vs. single CubeSat.
Software Development	$2,000,000	$2,000,000	Comparable software development costs, similar complexity.
Testing and Quality Assurance	$2,000,000	$2,500,000	Scaled up for larger number of satellites.
Manufacturing Costs	$3,000,000	$10,000,000	Larger scale manufacturing for 30 satellites.
Launch Costs	$1,000,000	$16,500,000-$67,000,000	Increased due to multiple launches and dedicated station launch.

Operational Costs	$1,500,000	$10,000,000	Includes ground stations and ongoing operational support.
Regulatory and Compliance Costs	$500,000	$2,000,000	Higher due to larger scale and international compliance needs.
Total Estimated Cost	**$18,000,000**	**$134,000,000**	**Reflects scaled-up project scope, complexity, and infrastructure.**

REVENUE GENERATION

Subscription Services:

- Premium access for advanced features such as enhanced analytics, detailed mission planning tools, and extended historical data.

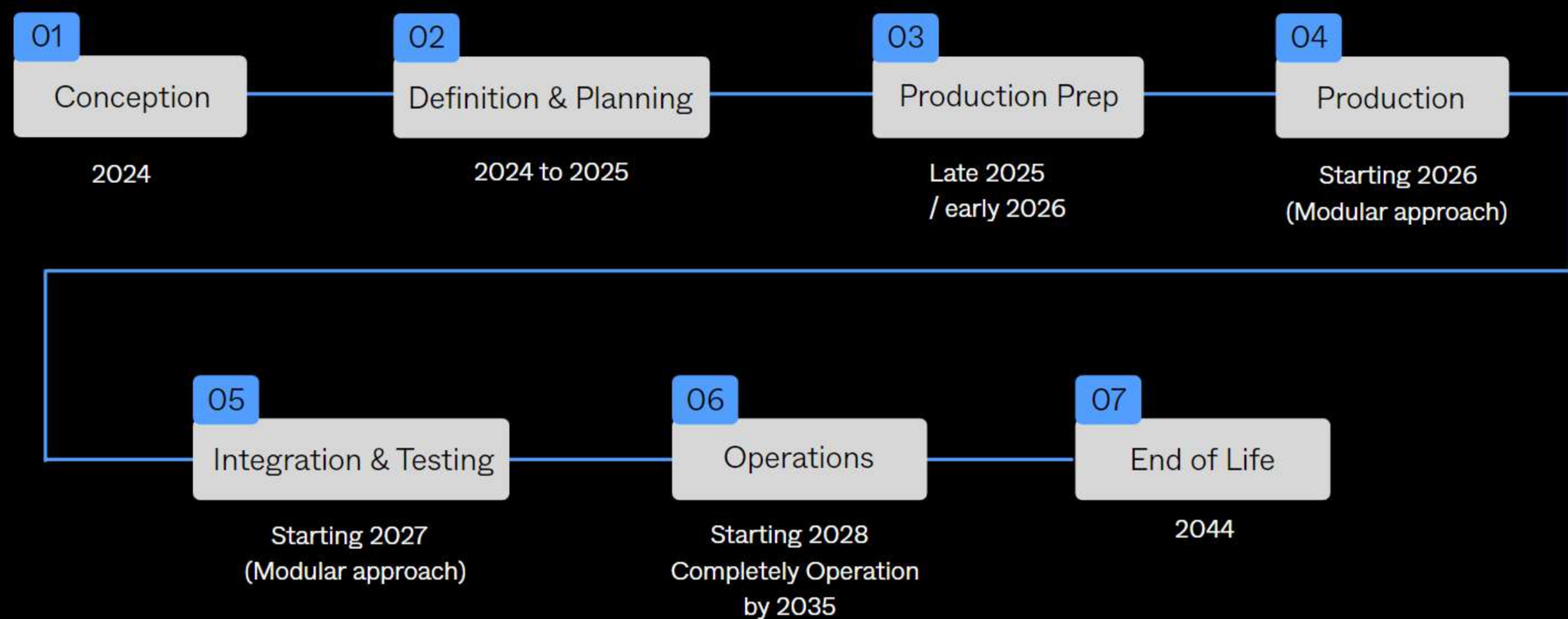
Custom Mission Support:

- Tailored services for specific mission requirements, including customized data analysis, orbital predictions, and dedicated support.

Satellite Deorbiting Services:

- Planned for future - Fee-based services for initiating deorbiting maneuvers for end-of-life satellites, managed through the MCC platform.

HIGH LEVEL PROJECT SCHEDULE



THE TEAM

Leadership Team

Cornelia Kairat & Sara Lehmann

Project Management, Mission Directors

Technical Team

Mario Tejedor Serrano, Tobias Hilgeroth, Ratan Purohit

Engineers, Data scientists, satellite operations specialists

Finance & Marketing

Emilio Santamaria

Business plan, Marketing strategies

THANK YOU

STARTEC

SPACE TRASH AVOIDANCE AND
REMOVAL TECHNOLOGY ECOSYSTEM